

Customer Segmentation & Revenue Forecasting



INTRODUCTION

- The dataset spans 2,000 customers and 50,000 purchases over 24 months (Jan 2024 - Dec 2025)
- Customer Segmentation: Who are our customers, and how much are they worth?
- Revenue Forecasting: What will revenue do over the next 6 months, and how much can that forecast be trusted?



PROJECT OBJECTIVE

- Move beyond the company's existing labels and segment customers by actual behavior (RFM + K-means clustering)
- Estimate long-term Customer Lifetime Value and identify at-risk customers for win-back action
- Build an honest, testable revenue forecasting model using a trend + seasonal approach, validated on held-out data
- Forecast revenue for January-June 2026 with a realistic confidence range

Dataset Overview

- 40 fields falling into 5 groups: transaction identifiers, customer information, product information, transaction financials, sales channel & campaigns
- Customer Segmentation used: customer_id, transaction_date, net_revenue, customer_since — aggregated to one row per customer
- Revenue Forecasting used: transaction_date, net_revenue — aggregated to one total per calendar month
- Product, campaign, channel, and pricing detail fields were not used in this analysis but remain available for future work

CUSTOMER ANALYTICS

Key Questions Addressed

- Who are the most valuable customers?
- Are the natural customer segments reliable?
- What is each customer's future value (CLV)?
- Which inactive customers should be targeted for re-engagement?

Transaction Data



Customer Analysis



Segmentation



Business Insights



RFM FRAMEWORK

Measuring Customer Value

RFM evaluates customers using three metrics:

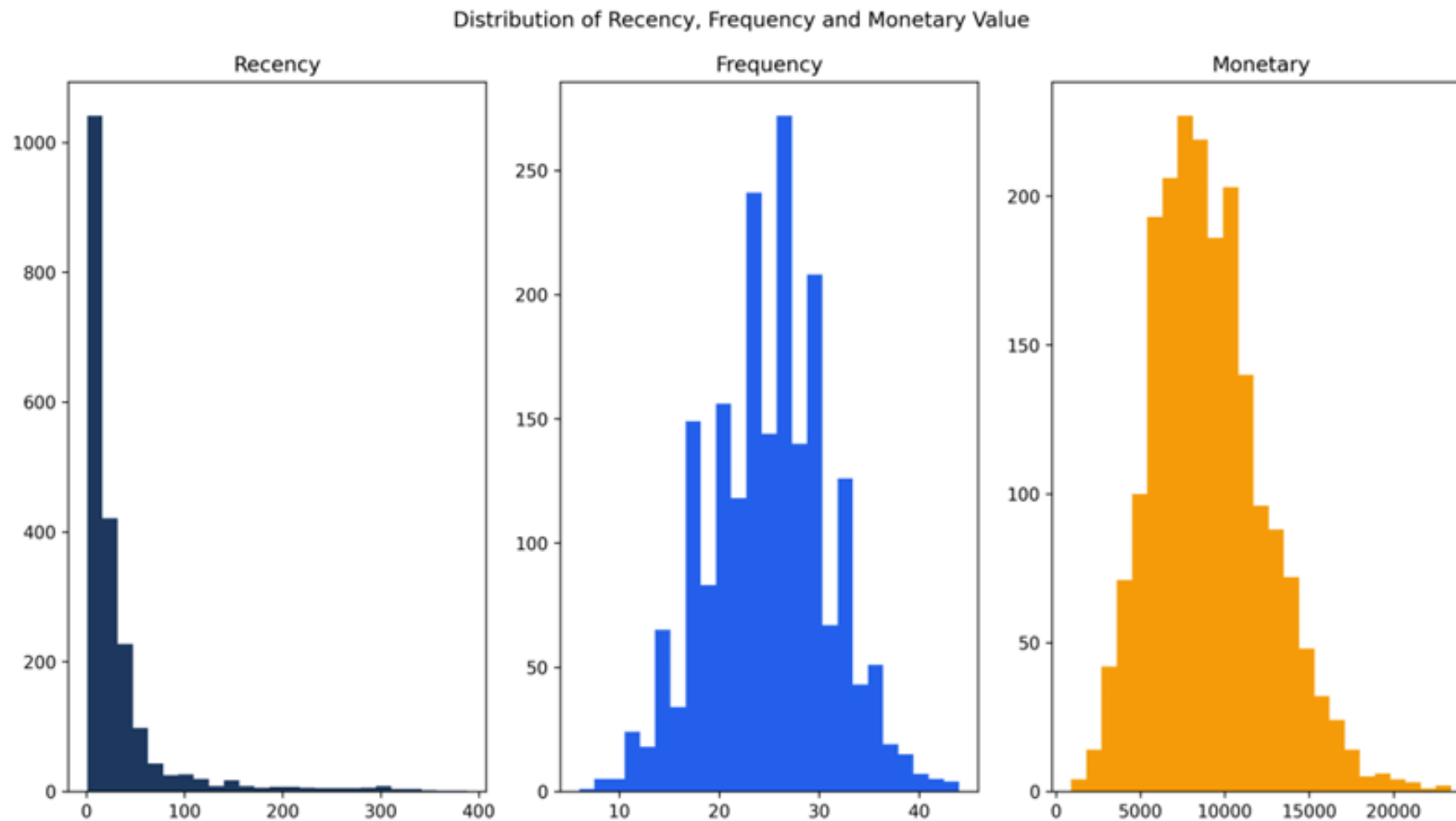
Metric	Meaning
Recency	Days since last purchase
Frequency	Number of purchases
Monetary	Total spending



Why RFM?

- Simple and widely used
- Identifies high-value customers
- Supports targeted marketing strategies

RFM SCORE DISTRIBUTION



- Recency is right-skewed, indicating many customers purchased recently.
- Frequency follows a bell-shaped pattern around the average number of purchases.
- Monetary values are concentrated at lower spending levels, with a few high spenders.

CUSTOMER SEGMENTATION USING RFM SCORES

Customers were grouped into three segments based on their RFM scores:

Segment	Customers
High Value	504
Medium Value	845
Low Value	651

- Medium-value customers form the largest segment.
- High-value customers represent the most profitable customer group.
- Low-value customers may require engagement strategies.

COMPARING ACTUAL BEHAVIOR WITH EXISTING CUSTOMER LABELS

The analysis compared behavioral segments with existing customer labels.

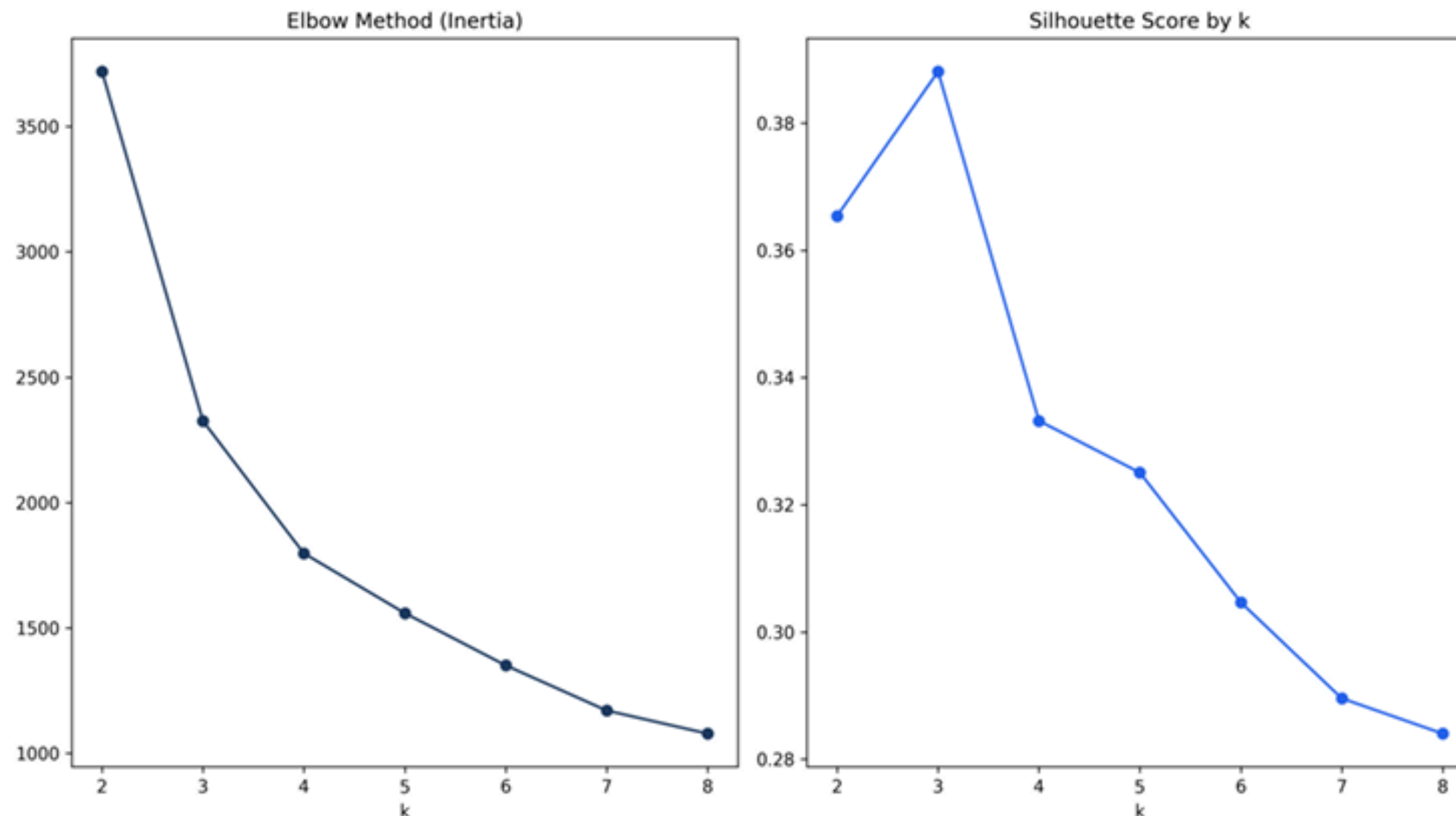
Behavior-based tier	Budget	New	Premium	Standard
High Value	131	75	73	225
Medium Value	207	119	128	391
Low Value	163	98	102	288

- 225 "Standard" customers were identified as High Value.
- 102 "Premium" customers were identified as Low Value.

Customer labels do not always reflect actual purchasing behavior.

CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING

Allow the data to identify natural customer segments without predefined rules.



Elbow Method

- Sharp reduction in within-cluster variation up to $k = 4$.

Silhouette Analysis

- Highest score at $k = 3$.
- $k = 4$ still provides acceptable cluster separation.

Four clusters were selected because they provide the best balance between Statistical quality, Business interpretability, and Practical usefulness

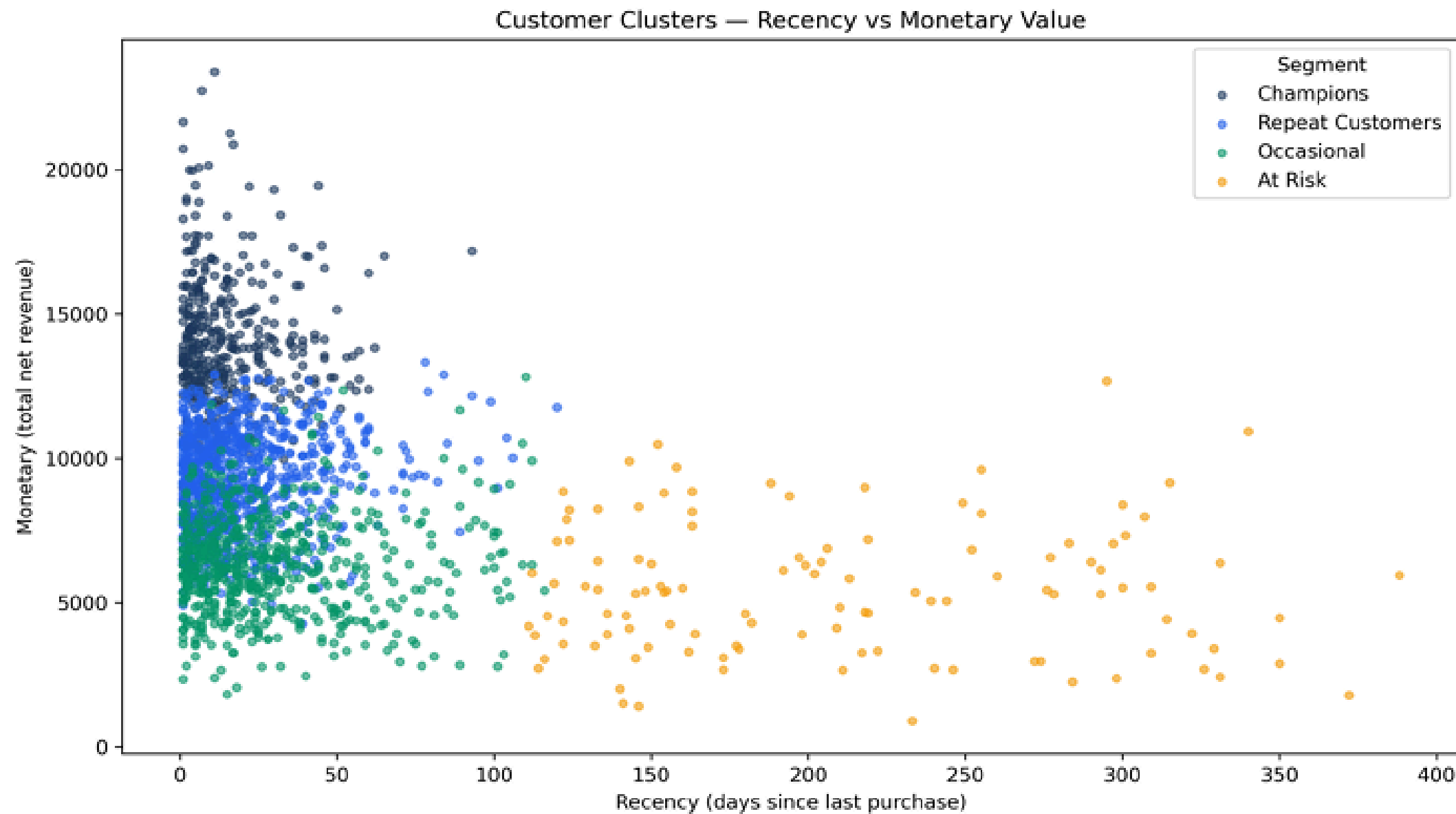
CUSTOMER SEGMENT PROFILES

Group	Customers	Avg. days since last purchase	Avg. purchases	Avg. total spend
Champions	376	15.6	31.6	LKR 14,072
Repeat Customers	896	18.4	27.1	LKR 9,258
Occasional	614	28.5	19.6	LKR 6,356
At Risk	114	209.1	15.9	LKR 5,481



- Champions purchase most recently and spend the most.
- At Risk customers purchase least frequently and have the lowest spending.

CLUSTER VISUALIZATION



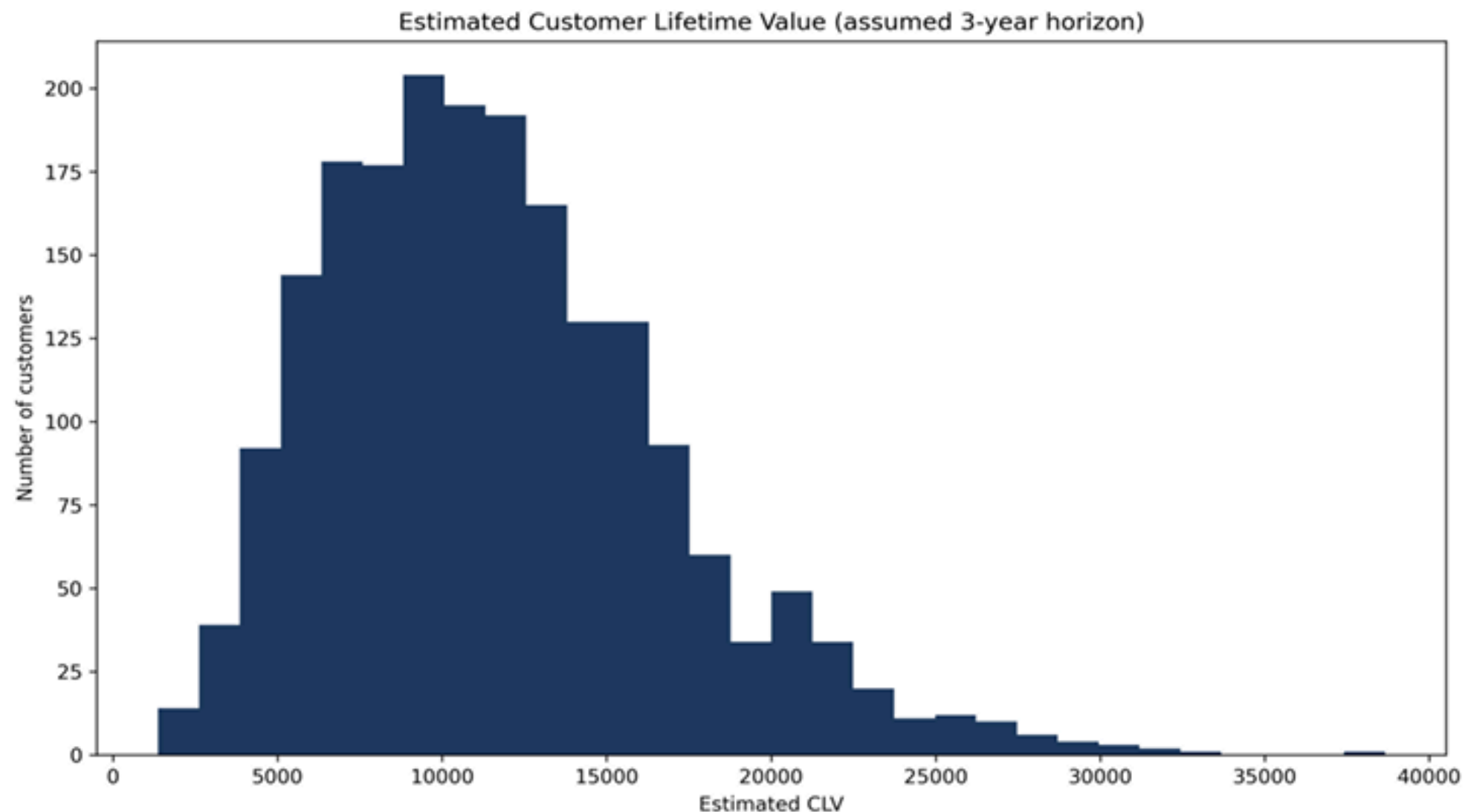
- Champions are concentrated at low recency and high spending.
- Repeat Customers occupy the middle region.
- Occasional Customers show moderate spending.
- At Risk Customers are located at high recency and low spending.

The four customer segments are well separated according to purchasing behaviour.

ESTIMATING CUSTOMER LIFETIME VALUE (CLV)

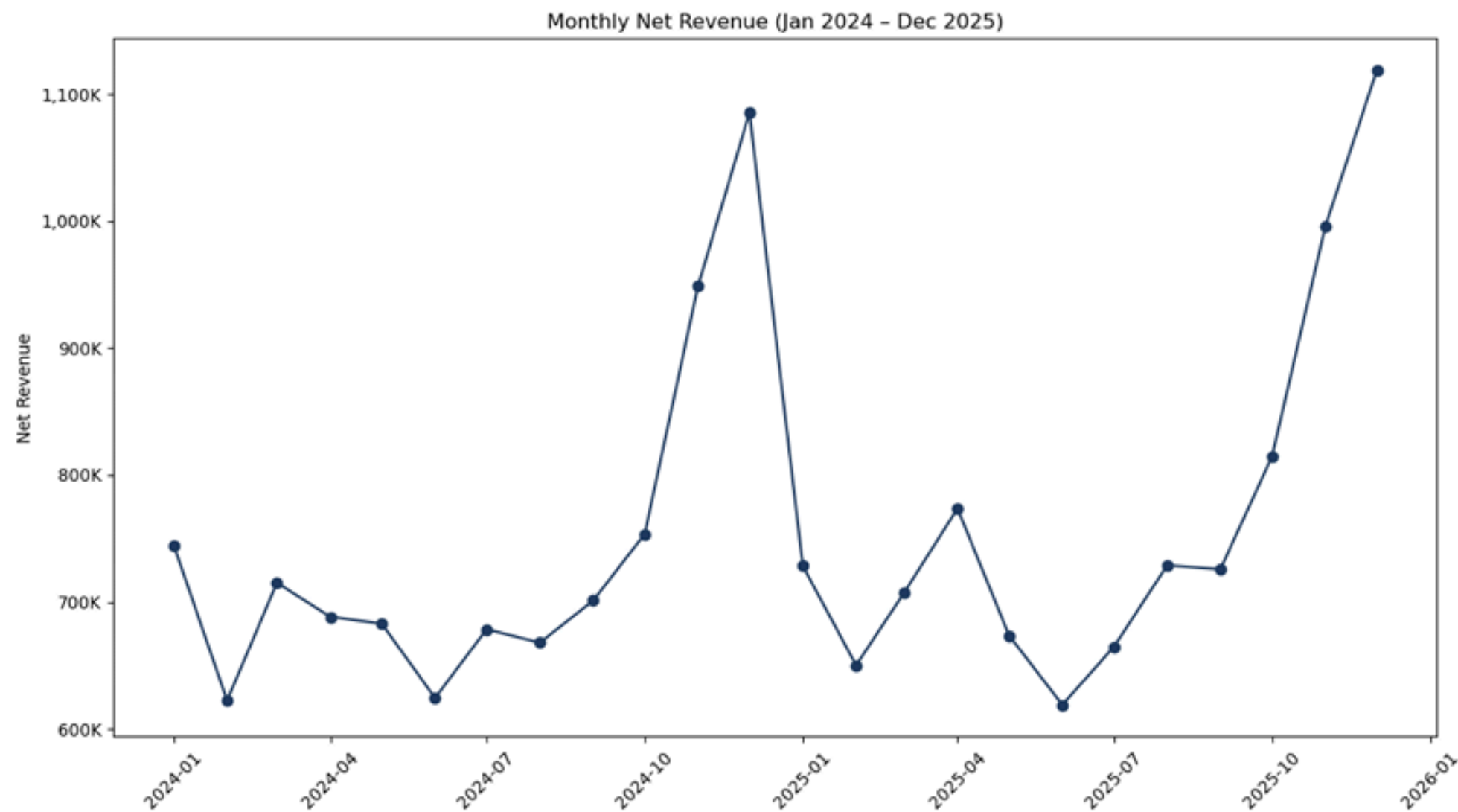
Estimate the long-term revenue generated by an average customer using historical purchasing behaviour.

$$\text{CLV} = \text{AVERAGE ORDER VALUE} \times \text{PURCHASE FREQUENCY} \times \text{CUSTOMER LIFESPAN}$$



- Most customers have an estimated CLV between approximately LKR 5,000 and LKR 15,000.
- A relatively small number of customers have substantially higher CLV, creating a right-skewed distribution with a long upper tail.

REVENUE FORECASTING

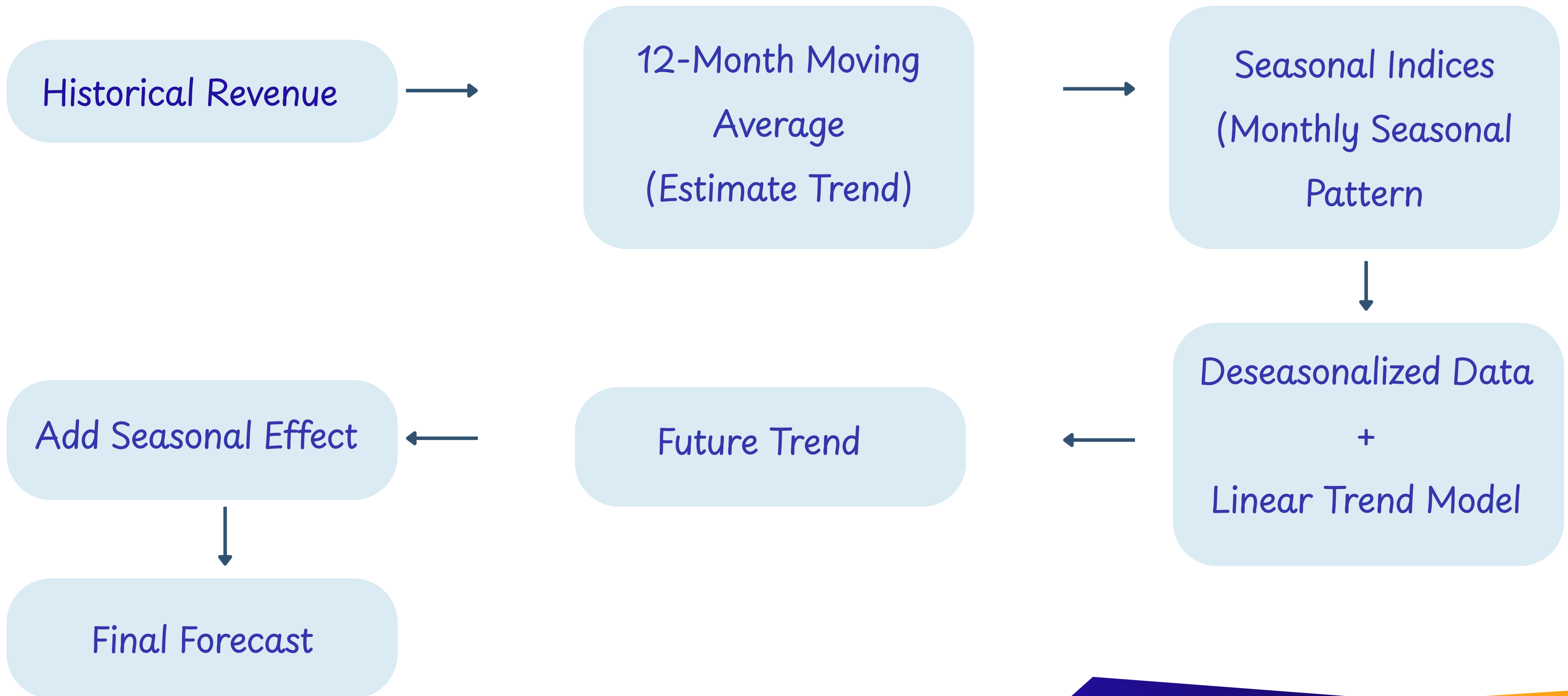


Classical Trend
+
Seasonal
Decomposition

Only 24 months of historical
data (2 annual cycles)
Insufficient data for reliable
complex seasonal models



FORECASTING METHODOLOGY



MODEL VALIDATION

- Last 3 months (Oct-Dec 2025) reserved as a holdout dataset
- These observations were not used when building the model
- Compared against a Naïve Flat Average Forecast



Training Data

Jan 2024 ----- Sep 2025 | Oct Nov Dec

Build

Test

The model was evaluated using unseen data to measure genuine predictive performance.

FORECAST ACCURACY

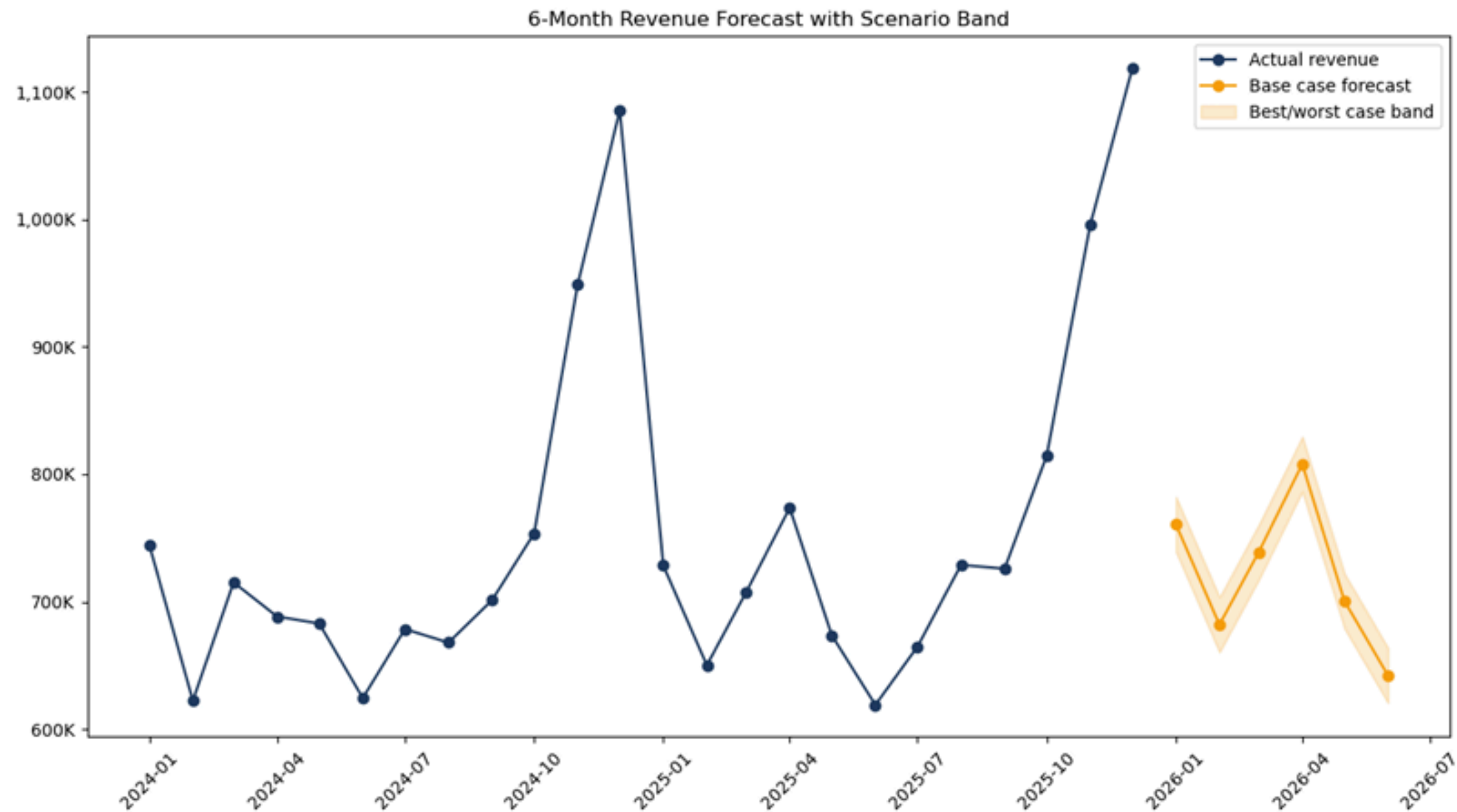


Model	MAE (LKR)	RMSE (LKR)	MAPE (%)
Naive (flat average)	269,784	297,288	26.38
Trend + Seasonal Index	18,167	22,150	2.06

- Reduced forecasting error from 26.4% to 2.1%
- Captured both trend and seasonal variation

The Trend + Seasonal model substantially outperformed the naïve baseline.

SIX-MONTH REVENUE FORECAST



- April expected to generate the highest revenue.
- June expected to have the lowest revenue.
- Forecast follows the historical seasonal pattern.

CONCLUSION

- Customer segmentation splits the base into High Value, Medium Value, and Low Value customers using RFM scoring
- The legacy labels frequently disagree with actual customer value
- K-means clustering confirmed four natural, well-separated customer groups
- Estimated Customer Lifetime Value for most customers falls between LKR 5,000-15,000
- The Trend + Seasonal forecasting model is highly accurate on unseen data
- The six-month forecast follows the historical seasonal pattern, with April expected highest and June expected lowest

RECOMMENDATIONS

- Prioritize the At-Risk cluster (114 customers) for a targeted win-back campaign
- Replace the Premium/Standard/Budget/New labels with the behavior-based RFM tier for service prioritization and targeted offers
- Use the Jan-Jun 2026 base-case forecast for operational planning, while keeping the worst-case band as a contingency floor
- Revisit both models once more transaction history becomes available



THANK YOU